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Dear Editors of *Blockchains*,

On behalf of my co-authors, I am pleased to submit our manuscript entitled **"Enhancing Consent Policy Privacy in Blockchain-Based Healthcare Data Sharing"** for consideration as an Article in *Blockchains*.

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Significance and context. Blockchain-based consent management promises verifiable and auditable enforcement of healthcare data-sharing policies, but public-chain transparency creates a privacy problem that existing work leaves open: the consent policies themselves. Prior privacy-preserving blockchain solutions protect data contents, user identities, or transaction details, and even hidden-policy encryption schemes do not control the policy-dependent artifacts—calldata layout, events, verifier interfaces, and gas usage—that a public chain exposes while enforcing a policy. This matters in healthcare, where a consent policy built from biomedical data-use terms (e.g., Data Use Ontology conditions) can itself reveal sensitive information about the data subject and the protected resource.

Our manuscript addresses this gap with TPS-ZKPI, a Trusted Proof Service-assisted zero-knowledge policy interpreter. It compiles heterogeneous Boolean consent policies into fixed-length programs evaluated by a single reusable zero-knowledge circuit, binds them to on-chain commitments, and normalizes the public-chain transcript so that verification reveals minimal policy-dependent information. Policy confidentiality rests on an explicit, delimited off-chain trust boundary. Beyond the construction, the manuscript contributes an attack-driven empirical evaluation on the Polygon Amoy testnet: supervised and unsupervised metadata-only attacks target policy content, policy identity, and policy structure and parameters under a multi-seed permutation-test criterion. No modeled attack recovers hidden policy information reproducibly above chance, and a paired gas-padding ablation isolates randomized gas padding as the defense that closes the gas channel, at a measured overhead of about 38% on access and 40% on commit. To our knowledge, this is the first work to define and empirically test policy privacy for blockchain consent enforcement at the level of the public-chain transcript rather than the policy representation alone.

Fit to the journal. The manuscript falls squarely within the scope of *Blockchains*: it concerns privacy and security of smart-contract-based systems, zero-knowledge proofs on public blockchains, and a concrete healthcare data-sharing application, combining protocol design with a reproducible on-chain empirical evaluation.

Declarations.

- We confirm that this manuscript is original, has not been published elsewhere, and is not under consideration by any other journal; no part of its content is currently under consideration or published elsewhere.

- This manuscript has not previously been submitted to any MDPI journal.
- No identifying text has been removed from the manuscript: the submitted version contains the full author names, affiliations, and correspondence details.
- All authors have read and approved the submitted version and consent to its submission.
- Proposed and excluded reviewers, if any, will be provided in the submission system.

Thank you for considering our manuscript. We look forward to the reviewers' feedback.

Sincerely,

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(on behalf of all authors)
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